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10 kW three-phase hybrid inverters compared

Solar ESS product comparison · October 2026 · Solis, GoodWe, Deye, SRNE, Huawei

Our pick: Solis S6-EH3P10K02-NV-YD-L. The best-value 48 V three-phase hybrid in Thailand, with 220 A battery current, IP66, 2× surge for 10 s, unbalanced output on both ports, a generator port and dual backup ports. **Runner-up: GoodWe GW10K-ET-L-G10** where PV oversizing or a documented 10-year warranty matters more than price. Sungrow and Huawei take high-voltage batteries only, so they do not suit 48 V batteries.

	Solis S6-EH3P10K02-NV-YD-L	GoodWe GW10K-ET-L-G10	Deye SUN-10K-SG05LP3	SRNE HESP48100SH3	Huawei SUN2000-10K-MAP0 (HV)
Battery	48 V, 220 A	48 V, 220 A	48 V, 210 A	48 V, 220 A	600–980 V, LUNA only
Max PV input	16 kW	20 kW	16 kW	15 kW	18 kW
MPPTs / current	2 / 20 + 40 A	3 / 20 A each	2 / 26 + 26 A	2 / 22 + 22 A	2 / 16 A
MPPT range / max DC	200–850 V / 1000 V	150–850 V / 1000 V	200–650 V / 800 V	200–650 V / 800 V	160–1000 V / 1100 V
Backup / surge	10 kW / 2× 10 s	11 kVA / 20 kVA 10 s	10 kW / 2× 10 s	10 kW / 20 kVA	10 kW, needs SmartGuard
Max efficiency	97.6 %	97.8 %	97.6 %	97.5 %	98.6 %
IP rating	IP66	IP66	IP65	IP65	IP66
Generator input	10 kW	20 kVA	Shared gen/ smart-load port	Yes	No
Monitoring	SolisCloud	SEMS	DeyeCloud / SolarMan	Optional Wi-Fi	FusionSolar
Warranty (Thailand)	Confirm with distributor	10 yr	10 yr	5 yr	10 yr

Best in each area

Area	Winner
Value (48 V)	Solis
PV oversizing	GoodWe, 20 kW on 3 MPPTs
Backup power	GoodWe, 11 kVA continuous, 20 kVA generator input
Efficiency	Huawei 98.6 % (HV only); GoodWe 97.8 % among 48 V units
Warranty	GoodWe, 10 years documented

Area	Winner
Parallel / battery choice	Deye, up to 10 units, widest battery list

Why Solis S6 stands out (10 kW LV model)

Feature	What it means	S6-EH3P10K02-NV-YD-L
High surge in backup	Starts pumps, air conditioners and motors without tripping	2× rated power for 10 s
High DC/AC ratio	More panels per inverter; surplus to battery or loads	Up to 16 kW PV (160 %)
Generator port	Generator wired straight in, auto start/stop, peak shaving	10 kW generator input
Dual backup outputs	Critical and non-critical loads on separate ports	Yes
High-current inputs	Suits 600 W+ panels and large batteries	20/40 A per MPPT; 220 A battery
Fast UPS switchover	Electronics keep running in a blackout	Under 10 ms (Solis S6 claim; confirm on LV datasheet)
Unbalanced output	Uneven loads across three phases	Grid and backup ports

Specifications are taken from the manufacturers' published datasheets; check the current datasheet before ordering. For prices and a design for your site, book a consultation at www.solaress.org or LINE: solaress1.

Sources: Solis datasheet · GoodWe ET-LV · SRNE datasheet · Huawei MAP0 · Solis S6 overview

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